ISD316D – PHASE 1
INTERNFLOW PROJECT BRIEF

Project Title
InternFlow (Pty) Ltd
A Scalable Multi-Tenant Internship & Learnership Management Infrastructure for South Africa

Duration
14 Weeks
Team Size
10 Students
Methodology
Full Systems Development Life Cycle (SDLC) Implementation

Project Context
InternFlow (Pty) Ltd is a private technology company developing a scalable digital infrastructure platform designed to manage internships, learnerships, mentorship programs, and youth development initiatives across South Africa.
Currently, most training providers operate using fragmented systems:

- Manual spreadsheets
- WhatsApp communication chains
- Paper-based onboarding
- Email-based document exchange
- No centralized compliance tracking
- Weak monitoring of learner progression
- Poor document retention practices

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These inefficiencies create:

- Compliance risk (SETA / CETA requirements)
- Administrative overload
- Data security vulnerabilities
- Limited scalability
- Poor reporting visibility
- Stipend tracking complications

InternFlow is building a **multi-tenant cloud-based infrastructure** to centralize and modernize this environment.
You have been contracted as an external Systems Architecture & Informatics Consulting Team to:

- Analyse the business case
- Validate system feasibility
- Design a scalable architecture
- Produce enterprise-level documentation
- Evaluate national rollout viability

This is not a coding assignment.
This is a full SDLC-based systems analysis and deployment planning engagement.

Strategic Positioning

This is a private-sector national-scale digital infrastructure simulation.
InternFlow intends to:

- Onboard 100+ training providers
- Support thousands of coordinators
- Manage millions of youth records
- Operate under POPIA compliance
- Integrate AI-based document validation
- Scale across rural and urban environments

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Your design must survive real-world scrutiny.

Core System Vision

InternFlow consists of three architectural layers:

1. Platform Governance Layer (InternFlow Head Office)

- Tenant approval & onboarding
- Compliance oversight
- Risk monitoring
- Sales & stakeholder management
- National analytics

2. Tenant Layer (Training Providers / Organisations)

Each tenant:

- Manages its own students
- Configures its own compliance rules
- Tracks onboarding & documentation
- Monitors logbooks & performance
- Generates reporting packs
- Manages coordinators

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3. Student Layer (Participants)

Students:

- Apply for opportunities
- Upload documents
- Complete onboarding
- Submit logbooks
- Track progression
- Download payslips
- Communicate with coordinators
- Receive certificates

This is a **true multi-tenant architecture**.

Objectives

Your team must:

1. Execute all SDLC phases
2. Validate business viability
3. Produce enterprise-grade documentation
4. Design for national scalability
5. Model governance & compliance structures
6. Prepare for simulated executive defense

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Mandatory Team Structure
Each student must assume a defined role.

Executive & Governance Layer

1. Project Manager
2. Business Analyst
3. Compliance & Risk Lead

Technical Architecture Layer

4. Solution Architect
5. Data Architect
6. Security Architect

Engineering & Modelling Layer

7. Systems Analyst (UML Lead)
8. Backend Architecture Lead
9. Frontend / UX Strategy Lead
10. DevOps & Deployment Lead

Each role must produce formal documented outputs.
No duplicated responsibilities.
Clear accountability is required.

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SDLC Phases

Phase 1: Initiation & Feasibility (Weeks 1–2)

Deliverables:

- Business Case Validation
- Market & Competitor Analysis
- Stakeholder Identification
- Feasibility Study:
  - Technical Feasibility
  - Operational Feasibility
  - Economic Feasibility
  - Legal Feasibility (POPIA)
- Project Charter
- Scope Definition
- Risk Pre-Assessment

Gate Review 1:

InternFlow Executive Approval to Proceed

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Phase 2: Requirements Engineering (Weeks 3–4)

Deliverables:

- Stakeholder Interview Simulation
- Functional Requirements Specification
- Non-Functional Requirements Specification
- Multi-Tenant Requirements Model
- Compliance Requirements Framework
- Use Case Catalogue
- Requirements Traceability Matrix

Gate Review 2:

Requirements Sign-Off

Phase 3: System Analysis & Logical Design (Weeks 5–6)

Deliverables:

- Use Case Diagrams (Expanded)
- Activity Diagrams
- Sequence Diagrams
- Domain Model
- Initial Data Model
- Tenant–User–Student relationship mapping

Focus:

Logical integrity before technology selection.

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Phase 4: Architectural & Physical Design (Weeks 7–8)

Deliverables:

- Component Diagram
- Deployment Diagram
- Database Schema Design
- RBAC Model (Role-Based Access Control)
- Authentication Design
- Tenant Isolation Strategy
- API Blueprint
- Infrastructure Blueprint

Architecture must address:

- Multi-tenant data separation
- High concurrency
- Rural bandwidth constraints
- Low-end device support
- Security layers
- Future AI integration
- Horizontal scalability

Gate Review 3: Architecture Approval

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Phase 5: Development Strategy & Prototype Simulation (Weeks 9–11)

Deliverables:

- High-fidelity UI Wireframes
- Multi-tenant Dashboard simulation
- Student lifecycle workflow
- Compliance engine mock structure
- AI document validation concept
- Data validation framework
- Logging & Monitoring model

Demonstrate:

- Offline-first logic or sync strategy
- Document validation logic
- 90-day certification tracking rule
- Fraud detection design approach

Phase 6: Testing & Quality Assurance (Weeks 12–13)

Deliverables:

- Test Plan
- Unit Test Strategy
- Integration Testing Framework
- Security Testing Strategy
- POPIA Compliance Testing Criteria
- UAT Scenarios
- Performance & Load Considerations
- Scalability threshold modelling

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Phase 7: Deployment & Maintenance Strategy (Week 14)

Deliverables:

- Deployment Strategy
- CI/CD Concept
- Tenant Onboarding Model
- Rollout Roadmap
- Disaster Recovery Plan
- Monitoring & Governance Framework
- Maintenance & Support Model

Final Output:

Executive Presentation & Technical Defense

Predictive Analytics Extension

InternFlow plans to integrate an AI-powered module for:

- At-risk learner detection
- Document fraud detection
- Performance pattern analysis
- Stipend anomaly detection

Your team must:

- Define data sources
- Propose OLTP vs OLAP separation
- Design data warehouse model
- Identify bias risks
- Define AI governance controls
- Design monitoring framework

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Risk Governance Expansion

The Compliance & Risk Lead must produce a full Risk Register including:

- Technical failure risk
- Data breach risk
- Regulatory non-compliance
- Budget risk
- Stakeholder resistance
- Platform misuse
- Tenant-level fraud
- AI bias risk

Include:

- Likelihood × Impact matrix
- Mitigation strategy
- Monitoring ownership

Assessment Structure

Criteria | Weight
SDLC Execution Integrity | 20%
Architectural Depth | 20%
UML & Technical Accuracy | 15%
Security & Compliance Design | 15%
Risk & Governance Framework | 10%
Innovation & Scalability | 10%
Executive Defense & Professionalism | 10%
Each role is assessed individually within team performance.

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Final Defense Scenario

You will present to a simulated InternFlow Executive Panel consisting of:

- Founder / CEO
- Chief Technology Officer
- Compliance Officer
- External Legal Advisor
- Financial Strategy Advisor

You must defend:

- Business sustainability
- Multi-tenant isolation
- Security maturity
- National scalability
- AI governance readiness
- Cost feasibility

If your architecture cannot survive enterprise-level questioning, it fails.

Strategic Reflection Layer

Before submission, answer:

- What breaks at 1 million users?
- What becomes legally sensitive?
- What data could expose InternFlow to liability?
- What redesign is required after year one?
- What must be automated before scaling?

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NDA Requirement

All students must sign a Non-Disclosure Agreement before receiving additional internal system documentation.
This is a private company engagement.
Intellectual property remains property of InternFlow (Pty) Ltd.

Closing Positioning

This project is not about drawing diagrams.
It is about designing real digital infrastructure for youth development in South Africa.
Design like it will scale.
Analyse like investors are watching.
Defend like compliance officers are auditing.

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